JAX-sklearn (xlearn) — Transparent Acceleration Architecture

User API, compatibility layer, JAX implementations, data policy, and hardware backends

Dashed frame = architecture zone · Solid frame = subsystem · Box = module and responsibility

PROJECT OVERVIEW

JAX-sklearn provides a drop-in scikit-learn API while transparently accelerating supported estimators with JAX. The architecture has a user API, a compatibility layer, JAX implementations, a data policy, and hardware backends.

USER CODE · sklearn-compatible API

- **import xlearn as sklearn** — 100% call surface
- **Estimators** — linear_model · cluster · decomposition · preprocessing
- **Calls** — fit · predict · transform

COMPATIBILITY · xlearn/__init__.py

- **__getattr__** — lazy submodule import
- **_auto_jax_accelerate_module** — replace estimator classes
- **AcceleratorRegistry** — _accelerator.py
- **create_intelligent_proxy** — _proxy.py
- **monkey_patch_estimator** — install proxy class

JAX ACCELERATION · xlearn/_jax

- **EstimatorProxy** — fit-time dispatch
- **create_accelerated_estimator** — _accelerator.py
- **_universal_jax.py** — JIT · vmap
- **original_class** — sklearn fallback
- **_gp.py** — Gaussian process
- **_mlp.py** — neural net
- **_svm.py** — SVM kernels
- **_tree.py** — tree models

DATA MANAGEMENT · arrays and policy

- **_data_conversion.py** — NumPy ↔ JAX · dtype · shape
- **_config.py** — enable_jax · jax_platform
- **jax_auto_threshold** — complexity >= 1e8
- **detect_hardware** — install command · platform probe

HARDWARE · JAX extras

- **CPU** — jax-cpu · jaxlib
- **CUDA GPU** — jax-gpu · jax-cuda11
- **TPU** — jax-tpu
- **Apple Metal** — jax-metal · float32 · experimental

The compatibility layer places EstimatorProxy between the sklearn API and either a JAX implementation or the original estimator.